SMARTGRAMA: SECURE MICROFINANCE AND WELFARE MANAGEMENT PLATFORM USING BLOCKCHAIN AND AI

COMMON INTEGRATED SOLUTION REPORT

Team member	Student ID	Individual component
Dissanayaka D M Y R	IT22145048	Permissioned blockchain loan verification
Jayasinghe A.I	IT22552310	Bilingual RAG conversational assistant
Muthukumarana H T D E	IT22111074	Machine-learning credit-risk and affordability support
Tharushika A J M H	IT22194626	Welfare eligibility, identity/KYC and audit workflow

Report submitted in partial fulfilment of the requirements for the
Bachelor of Science (Honours) in Information Technology

Department of Information Technology
Sri Lanka Institute of Information Technology
Sri Lanka

October 2026

DECLARATION AND SUPERVISOR CERTIFICATION

Candidate declaration. We declare that this common report is our own work and does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of our knowledge, it contains no material previously published or written by another person except where acknowledgement is made in the text. We grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this report, in whole or in part, in print, electronic, or other media, while retaining the right to use the content in future work.

Academic integrity and AI disclosure. The team must disclose generative AI and other automated tools in accordance with the applicable SLIIT policy. The authors remain responsible for the accuracy, originality, citations, code, data, analyses, and conclusions in this report.

Student name	Student ID	Signature	Date
Tharushika A J M H	IT22194626		
Dissanayaka D M Y R	IT22145048		
Muthukumarana H T D E	IT22111074		
Tharushika A J M H	IT22194626		

Supervisor certification. The above candidates carried out this research project under my supervision.

Role and name	Signature	Date
Supervisor: [Name]		
Co-supervisor: [Name]		

ABSTRACT

Microfinance and welfare administration can be constrained by fragmented institutional records, inconsistent appraisal practices, language barriers, and limited transparency over eligibility decisions. SmartGrama is a proposed integrated research platform that brings together four complementary components: a permissioned blockchain layer for cross-institutional loan verification; a Sinhala–English Retrieval-Augmented Generation (RAG) assistant for programme information; a machine-learning module for repayment-risk and affordability support; and a configurable welfare eligibility workflow using six Proxy Means Testing dimensions, authorised officer review, and audit records. The platform is designed around separation of concerns: conversational responses provide guidance, the risk model estimates patterns in historical records, the welfare engine applies configured rules, and the ledger records selected verification events. Detailed personal records should remain in protected off-chain storage, while only minimum necessary references or integrity commitments are considered for ledger recording. The source thesis draft reports a two-organisation Hyperledger Fabric loan-verification prototype, qualitative bilingual RAG tests with documented failure cases, 64.77% Random Forest accuracy with a high-risk one-versus-rest area under the curve (AUC) of 0.89 on 2,344 anonymised loan records, and 64 of 64 automated welfare-engine tests passed. The welfare test evidence described in that draft used a mock ledger provider; a subsequent team-reported live Fabric integration must be documented with its own reproducible evidence before submission. Results support prototype-level feasibility, not production readiness, social impact, or generalisable performance. The report consolidates the shared problem, architecture, integration boundaries, implementation evidence, evaluation limitations, governance needs, and a staged path toward controlled institutional validation.

Keywords: SmartGrama, Hyperledger Fabric, microfinance, welfare eligibility, retrieval-augmented generation

ACKNOWLEDGEMENTS

The team should replace this section with an accurate acknowledgement of the project supervisor, co-supervisor, participating stakeholders, technical contributors, and any institution that provided data or facilities. Do not name individuals or organisations unless their contribution is confirmed and acknowledgement is appropriate.

Table of Contents

DECLARATION AND SUPERVISOR CERTIFICATION 2

ABSTRACT 3

ACKNOWLEDGEMENTS 4

TABLE OF CONTENTS ..... **Error! Bookmark not defined.**

LIST OF FIGURES 8

LIST OF TABLES 8

LIST OF ABBREVIATIONS 9

1 INTRODUCTION AND PROJECT CONTEXT 10

1.1 Background and context 10

1.2 Stakeholders and user needs 10

1.3 Alignment with the Sustainable Development Goals 11

1.4 Purpose and scope of the common report 11

1.5 Report conventions and evidence boundary 11

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS 12

2.1 Overall problem definition 12

2.2 Root causes and overall research gap 12

2.3 Subproblem and component mapping 12

2.4 Overall aim and group objectives 13

2.5 Scope, assumptions and constraints 13

3 OVERALL SOLUTION STRUCTURE 15

3.1 Integrated solution overview 15

3.2 High-level architecture 15

3.3 Component summaries 16
3.3.1 Component 1 — Permissioned blockchain loan verification 16
3.3.2 Component 2 — Sinhala–English RAG assistant 16
3.3.3 Component 3 — Credit-risk and affordability support 17
3.3.4 Component 4 — Welfare eligibility, identity/KYC and audit workflow 17
3.4 End-to-end user and data flow 17
4 COMPONENT INTEGRATION AND COHESIVE OPERATION 19
4.1 Integration strategy 19
4.2 Interfaces, data contracts and shared models 19
4.3 Shared non-functional requirements 19
4.4 Integrated workflow and exception handling 20
4.5 Team contribution and dependency matrix 20
5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION 22
5.1 Common technology stack and infrastructure 22
5.2 Development and integration process 22
5.3 Deployment and operational configuration 22
5.4 Implementation status and evidence boundaries 22
5.5 Evidence of team execution 23
6 INTEGRATED TESTING, EVALUATION AND VALIDATION 24
6.1 Evaluation framework and success criteria 24
6.2 Integration and end-to-end testing 24
6.3 Overall results 24
6.4 Model comparison 25
6.5 Stakeholder and user validation 25
6.6 Achievement of common objectives 25
6.7 Integrated limitations and threats 26

7 COMMERCIALISATION, SUSTAINABILITY AND RESPONSIBLE INNOVATION 27
7.1 Value proposition and beneficiaries 27
7.2 Market alternatives and differentiation 27
7.3 Business model and feasibility 27
7.4 Adoption, intellectual property and compliance 28
7.5 Sustainable Development Goal contribution 28
7.6 Deployment risks and mitigations 28
8 CONCLUSION AND FUTURE DIRECTIONS 29
8.1 Overall conclusion 29
8.2 Principal integrated contribution 29
8.3 Prioritised future directions 29
8.4 Final conclusion 30
REFERENCES 31
APPENDICES 33
Appendix A — Integrated interface specification 33
Appendix B — Welfare test summary reported in the source draft 33
Appendix C — RAG failure cases reported in the source draft 33
Appendix D — Team contribution declaration 34
Appendix E — Final submission verification 34

LIST OF FIGURES

Figure 3.1 High-level SmartGrama integrated architecture

Figure 4.1 End-to-end user and data workflow

LIST OF TABLES

Table 1.1 Stakeholders and needs

Table 2.1 Component-to-subproblem mapping

Table 2.2 Objectives and evidence

Table 3.1 Component summary

Table 4.1 Shared non-functional requirements

Table 4.2 Team contribution and dependency matrix

Table 5.1 Implementation status and evidence boundaries

Table 6.1 Reported quantitative results

Table 6.2 Integrated evaluation and remaining evidence

Table 7.1 Deployment risks and mitigations

LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
AUC	Area Under the Curve
DID	Decentralised Identifier
DLT	Distributed Ledger Technology
FAISS	Facebook AI Similarity Search
KYC	Know Your Customer
LLM	Large Language Model
ML	Machine Learning
NLP	Natural Language Processing
PII	Personally Identifiable Information
PMT	Proxy Means Testing
RAG	Retrieval-Augmented Generation
ROC	Receiver Operating Characteristic
SMOTE	Synthetic Minority Over-sampling Technique
ZKP	Zero-Knowledge Proof

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and context

Microfinance and social welfare services depend on trustworthy records, understandable programme information, consistent assessments, and accountable decisions. In settings where multiple institutions maintain separate systems, applicants may be asked to provide similar information repeatedly, while lenders may have an incomplete view of existing liabilities. Welfare applicants may also struggle to understand procedures when official information is difficult to access in their preferred language. These are the common operational concerns addressed by SmartGrama.

Digitalisation can reduce repetitive administration, but a digital platform does not automatically guarantee correctness or fairness. Centralised databases can concentrate access and control; distributed ledgers can preserve inaccurate submissions as well as valid records; language models can produce fluent but unsupported answers; and predictive models can reproduce limitations in their training data. The design therefore combines specialised components with explicit review, privacy, and evidence boundaries rather than treating automation as a substitute for institutional responsibility.

1.2 Stakeholders and user needs

Stakeholder	Primary need	Potential system support
Citizens / applicants	Understand requirements and submit information with fewer avoidable barriers	Sinhala–English guidance, clear status information, privacy-aware data handling
Microfinance institutions	Verify relevant existing liabilities and support consistent appraisal	Permissioned loan queries and advisory risk/affordability output
Welfare officers	Apply configured assessment rules and explain decisions	PMT score breakdown, evidence review, explicit approve/reject action
Programme administrators	Maintain auditable and versioned eligibility processes	Rule governance, event records, access control and exception reporting
System administrators / auditors	Protect systems and verify record integrity	Role-based access, controlled logs, hash verification and operational monitoring

These needs are design targets, not proof that every stakeholder has validated the platform. The source material does not establish a recruited end-user study or a

longitudinal deployment. Claims about satisfaction, reduced poverty, reduced default, or improved institutional efficiency must therefore remain prospective.

1.3 Alignment with the Sustainable Development Goals

The project is plausibly related to SDG 1 (No Poverty), SDG 8 (Decent Work and Economic Growth), SDG 9 (Industry, Innovation and Infrastructure), and SDG 16 (Peace, Justice and Strong Institutions). Its intended contribution is indirect: more accessible information, more verifiable institutional records, and more reviewable administrative workflows may support responsible service delivery. The prototype has not measured changes in poverty, access to credit, institutional trust, or public-sector effectiveness; no direct SDG impact should be claimed.

1.4 Purpose and scope of the common report

This report explains the shared problem and integrated SmartGrama solution. It summarises the four individual components, their interfaces, common data and security boundaries, implementation status, team-level evaluation, and shared contribution. Detailed algorithmic decisions and component-specific experiments belong in the individual theses. The report uses the available SmartGrama full-platform thesis draft as its primary project evidence and distinguishes reported evidence from subsequent team-reported implementation updates and recommendations.

1.5 Report conventions and evidence boundary

Statements are classified implicitly as: (i) reported evidence in the source thesis draft; (ii) implementation status subsequently reported by the team; or (iii) proposed future work. The final authors must verify the second category against repository commits, configuration, transaction records, screenshots, and test logs. No unreported metric, participant study, deployment scale, or official policy equivalence is assumed.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall problem definition

The common problem is how to coordinate microfinance and welfare-related workflows across institutional boundaries while improving record verifiability, access to understandable information, repayment-risk triage, and welfare-decision auditability without exposing unnecessary personal data or delegating consequential decisions entirely to automated systems.

2.2 Root causes and overall research gap

The project framing identifies four interrelated causes: (1) fragmented records and limited cross-institutional visibility of existing liabilities; (2) language and information-access barriers when citizens seek welfare or micro-loan guidance; (3) difficulty applying consistent, data-informed repayment-risk and affordability assessments; and (4) limited ability to reproduce and audit welfare eligibility decisions. These causes require different methods. A ledger cannot answer policy questions, a conversational assistant cannot establish loan truth, a classifier cannot determine statutory eligibility, and a rule engine cannot independently validate the truth of submitted evidence.

The research gap is therefore an integration gap rather than a claim that blockchain, RAG, machine learning, or PMT is individually novel. The intended contribution is a modular platform that defines how these different outputs can support a common workflow while preserving separate trust boundaries and human decision authority.

2.3 Subproblem and component mapping

Component	Subproblem addressed	Input and method	Output / contribution
1. Permissioned blockchain loan verification	Separate institutions may not share a consistent, verifiable liability view	Authorised loan metadata; Hyperledger Fabric and chaincode	Queryable, tamper-evident transaction history for recorded liabilities
2. Bilingual RAG assistant	Applicants may have difficulty locating or understanding reliable programme information	English/Sinhala query; multilingual embeddings, FAISS retrieval, translation and local LLM	Context-informed conversational guidance; not an official decision
3. Credit-risk and affordability support	Appraisal may rely on incomplete or inconsistent interpretation of borrower information	Anonymised loan records; engineered income/affordability features; Random Forest	Advisory risk category and affordability estimate

4. Welfare eligibility, identity/KYC and audit workflow	Configured welfare assessments require repeatable scoring, identity linkage, officer review and auditability	Applicant/household inputs; six PMT dimensions; SHA-256 integrity checks and ledger audit integration	Score breakdown, officer verdict and auditable event trail

2.4 Overall aim and group objectives

Aim: To design and evaluate an integrated research prototype that supports secure, understandable, and auditable microfinance and welfare workflows through permissioned record verification, bilingual information retrieval, risk-support analytics, and human-reviewed welfare assessment.

Objective	Evidence expected	Status based on available evidence
O1. Demonstrate permissioned cross-institutional loan record creation and query	Valid transaction response, query results, authorisation and latency logs	Prototype transaction/query evidence reported; verify against current repository
O2. Provide grounded Sinhala and English information access	Representative bilingual questions, retrieved passages, response review and failure records	Functional examples reported; no aggregate accuracy percentage supplied
O3. Evaluate credit-risk and affordability support	Held-out model metrics, class-specific evaluation and feature pipeline	2,344 records; 64.77% accuracy and class AUCs reported
O4. Support repeatable welfare scoring and human-authorised outcomes	Rule tests, lifecycle tests, hash mismatch tests, verdict controls and audit evidence	64 automated tests reported for prototype; live integration update requires separate evidence
O5. Define an integrated, privacy-aware workflow	Architecture, interface contracts, failure handling and security review	Architecture documented; complete end-to-end production-like validation not established

2.5 Scope, assumptions and constraints

- The platform is a research prototype for microfinance and welfare workflows, not a deployed national service.
- The loan-risk evaluation uses 2,344 anonymised records as described in the source draft; representativeness and external generalisation are not established.
- The RAG assistant supports Sinhala and English using a local LLM pipeline, but reported examples include unsupported or incorrect answers.
- The welfare engine uses six PMT dimensions and configurable thresholds. The source material does not provide all official weights and thresholds; official Aswesuma equivalence must not be asserted without verification.
- Detailed personal information should remain off-chain. Hashing supports integrity checking but is not encryption and does not itself anonymise data.

- The report does not claim a completed end-user field study, national-scale performance, production security certification, or measured social impact.

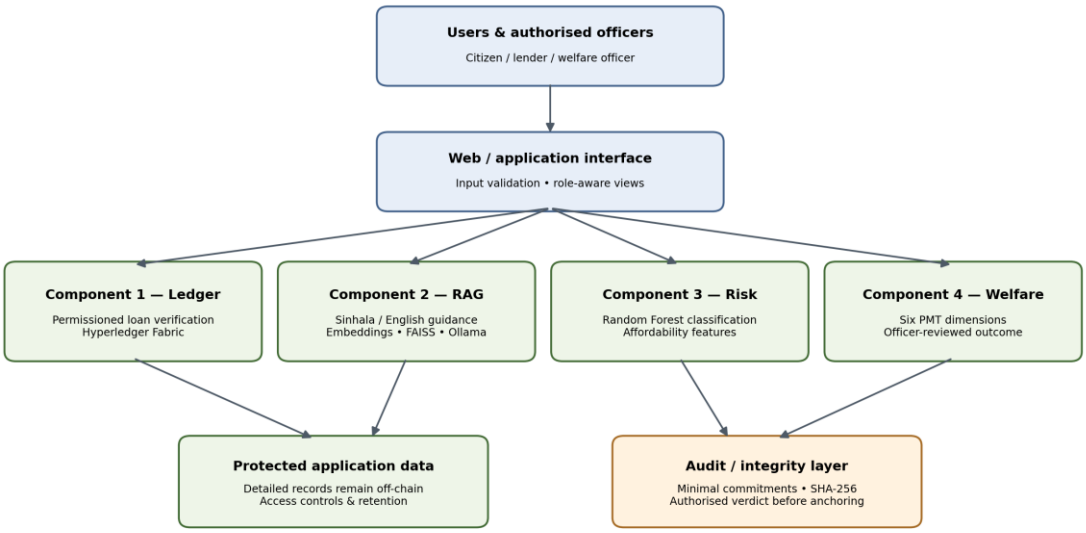
3 OVERALL SOLUTION STRUCTURE

3.1 Integrated solution overview

SmartGrama is structured as four specialised services behind a shared user-facing workflow. The ledger provides a verifiable record layer for permitted loan events; the RAG assistant retrieves relevant domain information before generation; the risk component estimates repayment-risk patterns and calculates an affordability-oriented indicator; and the welfare component computes a configured PMT score and supports an authorised officer's decision. A separate audit layer records minimum necessary evidence of outcomes and supports integrity verification.

Integration is achieved by defining what each output means and what it does not mean. A generated answer is guidance, a predicted risk class is an estimate, a PMT score is the result of configured rules, and a ledger event is evidence that a transaction was recorded under the network's rules. None independently proves that the underlying facts are true or that an applicant is legally entitled to a benefit or loan.

3.2 High-level architecture



Integrated target workflow; each component's implementation and evidence boundary must be stated separately.

Figure 3.1 High-level SmartGrama integrated architecture. The figure represents the intended logical architecture; implementation status and evidence differ by component.

The architecture separates the interface, application services, component-specific logic, protected application data, and the permissioned ledger. The ledger should contain only the data necessary for the agreed verification purpose. Access to detailed application documents, income evidence, guarantor information, household records, and other personal information should be controlled outside the ledger. The final report should align the diagram with the current implementation repository and mark any service that is not connected in the deployed build.

3.3 Component summaries

3.3.1 Component 1 — Permissioned blockchain loan verification

This component addresses cross-institutional loan visibility. The source draft describes Hyperledger Fabric with two organisations, a channel, JavaScript chaincode named loancc, and CreateLoan and QueryLoansByNIC operations. The design keeps detailed dossiers off-chain and places selected loan metadata or references on the ledger. A valid commit and sub-second query were reported for the described prototype. These observations demonstrate feasibility in that setup, not complete coverage of all lenders or proof that all submitted loan data are accurate.

3.3.2 Component 2 — Sinhala–English RAG assistant

The assistant combines a React interface and Flask service with a multilingual Sentence-Transformers embedding model, FAISS similarity search, translation through deep-translator, and a locally hosted model through Ollama. The pipeline retrieves relevant knowledge-base context and supplies it to a language model. The source draft documents functional bilingual examples and failure cases, including unsupported URLs, an incorrect registration answer, and additional advice beyond expected evidence. No aggregate answer-accuracy score is available in the source material.

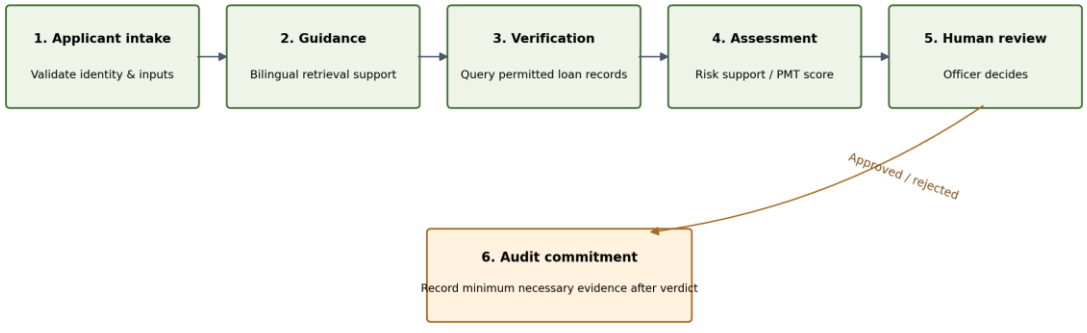
3.3.3 Component 3 — Credit-risk and affordability support

The risk component uses an anonymised dataset of 2,344 loan records, engineered income and affordability features, a stratified 80:20 split, SMOTE applied to the training subset, and a class-weighted Random Forest with 200 estimators. It reports 64.77% overall accuracy and one-versus-rest AUC values of 0.89 for High Risk, 0.79 for Low Risk, and 0.70 for Medium Risk. The result is advisory and requires further evaluation of per-class precision, recall, calibration, subgroup performance, and external validity.

3.3.4 Component 4 — Welfare eligibility, identity/KYC and audit workflow

The welfare component models six PMT dimensions: education; health and special needs; economic level and income; assets and ownership; housing conditions; and family demography. It maps a configured score to five named welfare classes and preserves an authorised officer as the final decision-maker. SHA-256 is used to verify canonical representations and detect changes. The source draft reports 64 automated tests across audit, eligibility, and lifecycle categories. The team has subsequently reported that live Hyperledger Fabric integration is completed; this report treats that as an implementation update to be substantiated by current code and live transaction evidence, while keeping the earlier mock-provider test results distinct.

3.4 End-to-end user and data flow



A failed service must be shown as unavailable—not interpreted as a negative verification result.

Figure 4.1 End-to-end user and data workflow. Audit recording follows an authorised verdict; failures must not be interpreted as successful verification.

A representative workflow begins with applicant intake and guidance. The application validates required fields and, where authorised, queries recorded loan liabilities. The risk component estimates risk and affordability using its defined inputs. Where welfare assessment is relevant, the welfare engine computes a score and presents a breakdown for officer review. The authorised officer records an explicit decision. A minimal audit event or commitment is then recorded, and integrity can be checked later. If the ledger is unavailable, the application must report that verification could not be completed rather than interpreting the failure as absence of a liability.

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration strategy

The integration strategy is interface-led and modular. Each component should expose a narrow contract, validate inputs independently, return explicit status and error information, and avoid sharing unnecessary personal data. The common application layer coordinates the sequence but does not collapse the component outputs into a single opaque score. Changes to chaincode, retrieval models, model features, or welfare thresholds should be versioned and tested separately before integration.

The common report distinguishes logical integration from a fully deployed end-to-end environment. The available thesis draft describes evidence from multiple component-specific environments. The subsequent team-reported live Fabric integration for the welfare audit path should be supported by repository references, configuration details, transaction IDs or sanitised logs, and repeatable test instructions. It should not retroactively relabel mock-provider tests as live-ledger tests.

4.2 Interfaces, data contracts and shared models

Interface	Minimum contract	Validation / failure rule
Loan record creation	Authorised institution; loan reference; type/volume; timestamp; minimum necessary metadata	Reject malformed or unauthorised requests; return transaction status; do not report a failed commit as success
Liability query	Authorised query reference and purpose; response limited to permitted fields	Distinguish no matching record from network, access, or query failure
RAG query	Question, language, request/session identifier as appropriate	Handle translation and retrieval failure explicitly; avoid unsupported procedural claims
Risk inference	Validated feature vector and model/version identifier	Reject missing/impossible values; return model version and advisory status
Welfare assessment	Validated household dimensions, rule-set version and score breakdown	Handle missing data and boundary conditions; do not silently apply unverified policy rules
Officer verdict and audit	Assessment reference, authorised officer, verdict, timestamp, canonical payload/hash	Block anchoring before verdict; log authorised changes; avoid storing full sensitive dossiers on-chain

4.3 Shared non-functional requirements

Requirement	Shared design expectation	Evidence / work still required

Privacy	Minimise collection and ledger disclosure; restrict access by role and purpose	Threat model, privacy impact review, retention rules and access-control tests
Integrity	Validate inputs, use canonical representations, check hashes and ledger transaction status	Independent verification, tampering tests and live integration evidence
Reliability	Make service status explicit and define retry/recovery behaviour	Fault injection, unavailable-service tests, backup/restore exercise
Usability and accessibility	Clear status messages, bilingual interaction and understandable score breakdown	Representative user testing and accessibility review
Explainability	Expose retrieved evidence, model advisory status and PMT score breakdown	RAG grounding review, model documentation and rule-version history
Maintainability	Version code, prompts, model artifacts, chaincode and policy rules	CI checks, reproducible setup and controlled release process
Performance	Measure latency and throughput under representative concurrent loads	Repeatable benchmark with hardware, sample sizes and percentile latency

4.4 Integrated workflow and exception handling

Failure handling is part of the business logic. A ledger timeout must be represented as unavailable verification, not as zero liabilities. An empty retrieval result must not be filled with invented policy details. A translation error should trigger clarification or a safe fallback. A model error or out-of-range input should not generate a confident risk category. Missing welfare fields should be surfaced for correction or manual review rather than silently replaced. Audit anchoring must remain blocked until an authorised verdict is recorded.

4.5 Team contribution and dependency matrix

The following mapping is a proposed allocation based on the four component areas in the project materials. Replace names and student IDs with the agreed team record and revise the entries to reflect actual contributions; do not use it as a substitute for evidence from version history, meeting records, or research diaries.

Team role / component	Primary responsibilities	Dependencies / shared work
[Student 1] — Blockchain	Fabric network, chaincode, loan creation/query, permissioning and ledger test evidence	Shared identifier and access model; integration and privacy review
[Student 2] — Bilingual RAG	React/Flask query flow, embeddings, vector search, translation, local LLM and answer review	Approved knowledge base, UI integration and safe-failure handling
[Student 3] — Credit-risk ML	Dataset preparation, feature engineering, split strategy, SMOTE, training and evaluation	Validated input contract, model deployment and risk interpretation
[Student 4] — Welfare / identity / audit	Identity/KYC workflow, six-dimension PMT scoring, officer lifecycle, SHA-256 and Fabric audit integration	Authorised rule set, ledger interface and audit lifecycle

All team members	Requirements, architecture, integration planning, common testing, documentation and presentation	Shared objectives, evidence consistency and release acceptance

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common technology stack and infrastructure

The documented technology choices are component-specific: Hyperledger Fabric and JavaScript chaincode for permissioned loan verification; React and Flask for the conversational assistant; Sentence-Transformers, FAISS, deep-translator and Ollama for retrieval and generation; a Random Forest model for credit-risk classification; and a TypeScript welfare engine for scoring and audit lifecycle logic. These choices reflect different computational needs and should be connected through explicit application interfaces rather than direct, undocumented dependencies.

5.2 Development and integration process

The project should be described as iterative research and development: define the common problem, design component responsibilities, implement and test individual components, define interfaces, integrate incrementally, and evaluate with component-appropriate evidence. Before submission, the team should insert authentic evidence of execution from repository history, issue tracking, meeting records, research diaries, or review notes. This draft does not invent sprint dates, commit counts, or team activity statistics.

5.3 Deployment and operational configuration

A repeatable deployment package should identify operating-system assumptions, dependency versions, environment variables, secret-management practices, network configuration, database setup, Fabric identities and certificates, model assets, vector-index creation, and test commands. The source draft does not provide every version or deployment command, so missing configuration must be retrieved from the actual project repositories. Secrets, personal records, and private keys must not be embedded in the submitted report.

5.4 Implementation status and evidence boundaries

Subsystem	Reported implementation/evidence	Boundary to preserve

Fabric loan verification	Two-organisation prototype; CreateLoan and QueryLoansByNIC; VALID commit and sub-second query reported	Does not prove national coverage, production throughput or complete source-data truth
Bilingual RAG	React/Flask application path; multilingual embeddings, FAISS, translation and Ollama; English/Sinhala examples	Known failure cases remain; no aggregate answer accuracy reported
Credit-risk model	2,344 anonymised records; 80:20 stratified split; SMOTE on training subset; Random Forest 200 estimators	64.77% accuracy is moderate; external validation and calibration not established
Welfare engine	Six PMT dimensions; officer verdict workflow; SHA-256 integrity checks; 64 automated tests passed; zero TypeScript errors reported	Earlier test suite used a mock ledger provider; current live Fabric integration must have separate evidence
Overall platform	Modular architecture and component workflows described	A single production-like end-to-end deployment and field trial are not established by the source draft

5.5 Evidence of team execution

For the final submission, attach a concise evidence summary rather than raw activity logs. Suitable evidence includes repository commit references for each component, approved architecture versions, test reports, interface review records, meeting decisions, and a contribution declaration agreed by all four members. Redact secrets, private keys, personal identifiers, and any sensitive borrower or welfare information.

6 INTEGRATED TESTING, EVALUATION AND VALIDATION

6.1 Evaluation framework and success criteria

Evaluation must match the question being tested. Ledger operations require transaction-validity, authorisation, query correctness and latency evidence. RAG requires separate measurement of retrieval relevance, answer correctness, language fidelity, groundedness and safe abstention. Credit-risk evaluation requires held-out predictive metrics and class-aware analysis. Welfare evaluation requires rule-boundary, lifecycle, hash-integrity and role-control tests. An integrated workflow additionally requires interface, failure-recovery, security and usability tests.

6.2 Integration and end-to-end testing

The source draft reports component-level results but does not establish a recruited participant study or a complete production-like end-to-end trial. The team should run a controlled end-to-end suite using synthetic data and record the code/configuration version, input, expected outcome, actual outcome, timestamp, and evidence location for every case. The live Fabric audit path reported by the team should be tested independently from the earlier mock-provider suite.

6.3 Overall results

Evaluation area	Reported result	Interpretation
Blockchain loan prototype	VALID commit; query latency reported as below one second	Evidence of transaction/query feasibility in the described test setup
RAG assistant	Representative bilingual operation; several failure cases documented	Functional feasibility only; answer quality needs a structured test set
Credit-risk model	Random Forest accuracy 64.77%; AUC: High 0.89, Low 0.79, Medium 0.70	Class separation varies; accuracy alone is insufficient
Welfare engine	64/64 tests passed: 17 audit, 28 eligibility, 19 lifecycle; zero TypeScript errors	Strong evidence for the tested logic and mock-provider suite, not by itself live-ledger correctness
Live welfare-to-Fabric integration	Subsequently reported as completed by the team	Insert authentic transaction logs, repeatable integration tests, error-path evidence and current code references before submission

6.4 Model comparison

The source draft reports Random Forest as the best of the compared models by overall accuracy. The exact accuracy figures for the two baseline models should be taken from the original research paper or actual evaluation output and verified before inclusion; they are not reproduced here to avoid transcription errors.

Model	Accuracy / metric available in current draft	Interpretation
Random Forest	64.77% accuracy; AUC High 0.89, Low 0.79, Medium 0.70	Reported leading model; class-level performance is uneven
Decision Tree	Baseline included in source comparison; exact value should be verified	Do not infer performance from the Random Forest result
Logistic Regression	Baseline included in source comparison; exact value should be verified	Needs the original results table for an exact numerical comparison

6.5 Stakeholder and user validation

The available project evidence does not report a recruited user study, formal stakeholder acceptance test, or field deployment. The team must not claim that citizens, lenders, or welfare officers validated the solution unless signed-off records exist. A future validation plan should recruit representative users with appropriate consent and institutional approval, evaluate bilingual comprehension and task completion, test officer interpretation of score breakdowns, and document concerns and changes.

6.6 Achievement of common objectives

Objective	Assessment	Justification
O1 — Permissioned verification	Partly supported	Prototype transaction/query evidence is reported; broader institutional coverage and current integration evidence must be documented
O2 — Bilingual information access	Partly supported	Functional examples exist, but failures are documented and no overall answer-accuracy metric is supplied
O3 — Risk and affordability support	Partly supported	A held-out model evaluation is reported, but accuracy is moderate and external validation/calibration are absent
O4 — Welfare scoring and auditability	Partly supported	64 tests passed in the described suite; live Fabric integration needs its own evidence and rule weights require verification
O5 — Integrated, privacy-aware workflow	Partly supported	Architecture and component boundaries are defined; full end-to-end

		production-like evaluation remains outstanding

6.7 Integrated limitations and threats

The principal threats are limited dataset representativeness, potential class imbalance, absent external model validation, qualitative rather than quantitative RAG evaluation, possible translation errors, unknown performance under concurrent load, incomplete privacy and threat-model evidence, and incomplete confirmation of the full integrated deployment. Immutable records can preserve inaccurate input; hashes do not conceal data; and automated scoring may produce unfair outcomes if rules or data are inappropriate. These limitations restrict the conclusions to prototype-level feasibility.

7 COMMERCIALISATION, SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value proposition and beneficiaries

The potential value proposition is a coordinated workflow that can reduce repeated information gathering, improve access to understandable guidance, support consistent risk triage, and make selected administrative events easier to verify. Potential beneficiaries include applicants, microfinance providers, welfare administrators, and auditors. These benefits remain hypotheses until tested in a controlled pilot using operational measures such as task completion, processing time, query failure rate, decision review effort, and user comprehension.

7.2 Market alternatives and differentiation

Alternative approaches include centralised loan databases, standalone welfare portals, conventional FAQ/chatbot systems, separate credit-scoring tools, and audit logs maintained within one institution. SmartGrama's intended differentiation is the combination of these capabilities with a permissioned ledger and human-reviewed welfare workflow. The architecture is not universally superior: a shared centralised service may be simpler where governance and trust arrangements already exist, while a ledger adds operational complexity. Differentiation should be validated through comparative requirements and pilot evidence rather than asserted as market leadership.

7.3 Business model and feasibility

Possible operating models include institution-funded deployment, a consortium-operated service, or a managed service provided under formal agreements. Feasibility depends on integration with existing institutional systems, ongoing support, Fabric identity and certificate administration, secure hosting, model and knowledge-base maintenance, staff training, and compliance review. The current research evidence does not provide a validated market size, willingness-to-pay study, cost model, or revenue forecast; financial projections should not be invented.

7.4 Adoption, intellectual property and compliance

Adoption barriers include trust between institutions, data-sharing agreements, integration costs, digital literacy, Sinhala-language quality, staff acceptance, and the need for accountable dispute and appeal mechanisms. The team should confirm software ownership, third-party licence obligations, dataset rights, institutional approvals, and permitted use of any identity or welfare data. A real deployment in Sri Lanka requires qualified legal and institutional review of applicable personal-data protection, financial confidentiality, welfare administration, retention, and cross-institutional sharing obligations.

7.5 Sustainable Development Goal contribution

The most plausible links are SDG 1 through the intended support for welfare access, SDG 8 through responsible access to microfinance, SDG 9 through digital infrastructure, and SDG 16 through accountable processes. Negative effects could include exclusion due to incorrect rules, inaccessible interfaces, biased risk predictions, translation errors, or unauthorised disclosure. Benefits depend on sustained governance, correction and appeal procedures, monitoring, and access for people with limited digital connectivity.

7.6 Deployment risks and mitigations

Risk	Potential consequence	Mitigation / evidence needed
Personal data exposed through ledger or logs	Long-lived disclosure and linkage risk	Data minimisation, privacy review, access controls, key management and payload inspection
Incorrect generated guidance	Applicants follow wrong procedures	Grounding tests, approved knowledge sources, safe abstention and human escalation
Biased or poorly calibrated risk model	Unfair appraisal or inappropriate affordability estimate	External validation, subgroup analysis, calibration and human review
Incorrect welfare weights or thresholds	Wrong classification or benefit recommendation	Authorised rule source, versioning, policy-owner approval and boundary tests
Fabric/network outage	Verification unavailable during application	Explicit unavailable status, retry policy, monitoring and recovery tests
Unclear accountability	Disputes cannot be resolved fairly	Named decision authority, appeals, audit review and incident procedures

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall conclusion

SmartGrama addresses a coherent cross-institutional problem through four complementary capabilities: permissioned loan verification, bilingual retrieval-grounded guidance, machine-learning risk support, and configurable welfare scoring with officer review and auditability. The reported prototype evidence demonstrates technical feasibility in several component-specific settings. It does not establish that the entire platform is production-ready, that its outputs are correct for all applicants, or that deployment would improve financial inclusion or welfare outcomes.

8.2 Principal integrated contribution

The principal shared contribution is an architecture that separates information retrieval, predictive risk estimation, deterministic welfare scoring, human authority, and ledger recording. This separation clarifies the meaning of each output and creates explicit places to apply privacy controls, failure handling, audit checks, and human review. The approach is valuable only if these boundaries are enforced in implementation and supported by evidence.

8.3 Prioritised future directions

1. Complete and publish reproducible end-to-end test evidence, including the reported live welfare-to-Fabric path, access-control checks, retry behaviour and negative cases.
2. Build a labelled English–Sinhala RAG evaluation set and measure retrieval relevance, factual correctness, groundedness, translation fidelity, abstention quality and failure rates.
3. Re-evaluate the credit-risk model using external or temporally separated data, per-class precision/recall/F1, calibration, subgroup checks and documented feature importance.
4. Verify welfare scoring weights and thresholds against an authorised, version-controlled specification; retain officer decision authority and provide appeal and correction mechanisms.

5. Complete privacy and security assessment for identity references, ledger payloads, logs, translation services, model prompts, keys and off-chain storage.
6. Conduct a supervised institutional pilot with clear acceptance criteria, participant consent where applicable, training, monitoring, incident response and a go/no-go review.

8.4 Final conclusion

The evidence supports continued controlled development of SmartGrama as an integrated research prototype. The platform's credibility depends less on the number of technologies combined than on the quality of its interfaces, the correctness of its data and rules, the transparency of its limitations, and the accountability of the people who use it. A deployment decision should be based on independently reviewable evidence rather than architectural intent or component-level success alone.

REFERENCES

[1] T. B. Brown et al., “Language Models are Few-Shot Learners,” Advances in Neural Information Processing Systems, vol. 33, pp. 1877–1901, 2020.

[2] T. Ranasinghe et al., “Sinhala Encoder-only Language Models and Evaluation,” Proc. 63rd Annual Meeting of the Association for Computational Linguistics, pp. 8623–8636, 2025. Verify metadata against the original publication.

[3] P. Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” Advances in Neural Information Processing Systems, vol. 33, pp. 9459–9474, 2020.

[4] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” Proc. NAACL-HLT, pp. 4171–4186, 2019.

[5] Y. Gao et al., “Retrieval-Augmented Generation for Large Language Models: A Survey,” arXiv:2312.10997, 2023.

[6] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks,” Proc. EMNLP-IJCNLP, pp. 3982–3992, 2019.

[7] J. Johnson, M. Douze, and H. Jégou, “Billion-scale similarity search with GPUs,” IEEE Transactions on Big Data, vol. 7, no. 3, pp. 535–547, 2021.

[8] N. Reimers and I. Gurevych, “Making Monolingual Sentence Embeddings Multilingual using Knowledge Distillation,” Proc. EMNLP, pp. 4512–4525, 2020.

[9] S. Nakamoto, “Bitcoin: A Peer-to-Peer Electronic Cash System,” 2008.

[10] E. Androulaki et al., “Hyperledger Fabric: A Distributed Operating System for Permissioned Blockchains,” Proc. 13th EuroSys Conference, 2018, doi: 10.1145/3190508.3190538.

[11] C. Cachin, “Architecture of the Hyperledger Blockchain Fabric,” Workshop on Distributed Cryptocurrencies and Consensus Ledgers, 2016.

[12] Z. Zheng et al., “An Overview of Blockchain Technology: Architecture, Consensus, and Future Trends,” IEEE International Congress on Big Data, pp. 557–564, 2017.

[13] F. Casino, T. K. Dasaklis, and C. Patsakis, “A systematic literature review of blockchain-based applications,” Telematics and Informatics, vol. 36, pp. 55–81, 2019.

[14] Hyperledger Foundation, “Hyperledger Fabric Documentation: Architecture and Components,” Linux Foundation, 2023. Verify the exact document title and date.

[15] G. Nanayakkara and J. Stewart, “Gender and other repayment determinants of microfinancing in Indonesia and Sri Lanka,” International Journal of Social Economics, vol. 42, no. 4, pp. 322–339, 2015. Verify bibliographic details.

[16] D. T. Priyankara and E. A. G. Sumanasiri, “Determinants of Microfinance Loan Default: An Empirical Investigation in Sri Lanka,” South Asian Journal of Social Studies and Economics. Verify volume, issue and pages.

[17] D. H. Wickramasinghe, “Study on Factors Influencing on Microfinance Loan Repayment,” International Research Journal of Advanced Engineering and Science, vol. 5, no. 4, pp. 125–129, 2020. Verify metadata.

[18] E. A. G. Sumanasiri and D. T. Priyankara, “Microfinance Loan Default in Sri Lanka: A Case Study of Matugama Division,” Insights into Economics and Management, vol. 1, pp. 118–127. Verify publication year and metadata.

[19] H. A. P. L. J. Hettiarachchi and R. Punchi-Manage, “Analysis of Loan Delinquency Prediction Based on Multinomial Logistic Regression and Random Forest,” IRC-OUSL 2025, 2025. Verify proceedings details.

[20] I. N. Manawadu et al., “Microfinance Interest Rate Prediction and Automate the Loan Application,” Proc. 2nd International Conference on Science and Technology, pp. 99–108, 2020. Verify metadata.

[21] National Institute of Standards and Technology, “Secure Hash Standard (SHS),” FIPS PUB 180-4, Aug. 2015.

[22] A. Sebastian et al., “A Proxy Means Test for Sri Lanka,” World Bank Policy Research Working Paper No. 8605, 2018.

[23] L. Breiman, “Random Forests,” Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.

[24] N. V. Chawla et al., “SMOTE: Synthetic Minority Over-sampling Technique,” Journal of Artificial Intelligence Research, vol. 16, pp. 321–357, 2002.

[25] T. Fawcett, “An introduction to ROC analysis,” Pattern Recognition Letters, vol. 27, no. 8, pp. 861–874, 2006.

APPENDICES

Appendix A — Integrated interface specification

Operation	Required request information	Successful response	Failure behaviour
Create loan record	Institution identity, authorised reference, permitted loan metadata	Transaction identifier and valid commit status	Explicit error; no success status on timeout or invalid transaction
Query loan records	Authorised query reference and purpose	Permitted liability records and status	Separate no-match response from access/network failure
RAG question	Question and language	Answer plus retrieved evidence where supported	Clarify, abstain, or refer to official source when evidence is insufficient
Risk inference	Validated feature vector and model version	Risk class and advisory affordability output	Reject invalid inputs and return a clear error
Welfare assessment	Six dimension inputs and rule-set version	Score, category and breakdown for officer review	Flag missing/conflicting data; require correction or manual review
Audit verification	Assessment reference and canonical representation	Hash comparison and recorded event status	Report mismatch or unavailable ledger distinctly

Appendix B — Welfare test summary reported in the source draft

Category	Count	Reported focus
Blockchain audit tests	17	SHA-256 canonical state hash, mismatch detection and minimised payload
Eligibility core tests	28	Calculations, threshold mapping and six PMT dimensions
Lifecycle tests	19	Identity bindings, specialised schemes and workflow transitions
Total	64	All tests passed; zero TypeScript compiler errors reported

Important: the source draft describes these tests as using a mock ledger provider. If the current implementation includes live Hyperledger Fabric integration, attach a separate live-integration test report. Do not conflate the two evidence sets.

Appendix C — RAG failure cases reported in the source draft

Case	Reported observation	Implication
Profile question	Relevant context retrieved, but unsupported URLs or steps were added	Retrieval relevance alone does not guarantee groundedness
NIC-only registration question	Generated answer incorrectly stated that registration was possible with NIC alone	Procedural claims require direct evidence and safe escalation
Sinhala repayment question	Additional loan-planning statements exceeded the expected evidence	Bilingual responses require groundedness and language-fidelity checks

Appendix D — Team contribution declaration

The four members should review and agree the component ownership, shared tasks, and evidence contributions recorded in Section 4.5. Insert verified names, student IDs, dates, and signatures according to university requirements.

Appendix E — Final submission verification

- Replace all student, registration, supervisor and date placeholders.
- Verify the current implementation status of each component against the source repository.
- Attach authentic evidence for the reported live welfare-to-Fabric integration, without inventing performance figures.
- Verify all reference metadata and ensure in-text citations map correctly to the final reference list.
- Confirm PMT weights and thresholds against the actual authorised rule configuration.
- Update the table of contents, lists, captions and page numbers in Microsoft Word.
- Check every figure and table is cited in the main text and numbered consistently.
- Remove this checklist if the department requires appendices to contain only research evidence.
- Disclose AI assistance in accordance with SLIIT policy and obtain supervisor review.